

1.16 Solution: The total energy, without the extra conductor, is given by

$$E_i[\psi_i] = \frac{1}{2} \int_V \nabla \psi_i \cdot \nabla \psi_i d^3x,$$

where ψ_i is the solution to the initial configuration as in Problem 1.15, and V is the volume exterior to the conductors. Now, introduce the extra conductor, which occupies a volume of V' , the total energy in the final configuration, assuming no change to the potential, is

$$E_f[\psi_i] = \frac{1}{2} \int_{V \setminus V'} \nabla \psi_i \cdot \nabla \psi_i d^3x,$$

i.e., the total energy calculated with the initial potential excluding the volume taken by the extra conductor. Obviously,

$$E_i[\psi_i] > E_f[\psi_i].$$